

Review

Current Progress Regarding *Cordyceps militaris*, Its Metabolite Function, and Its Production

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Abstract: *Cordyceps militaris* is a valuable medicinal fungus which has been widely used as a traditional medicine in East Asia. Compared to the well-known medicinal fungus *C. sinensis*, *C. militaris* can produce similar fermented metabolites with various biological activities, but it requires a shorter culture time and simpler culture conditions, and therefore, it has attracted increasing attention in recent years. The purpose of this review was to organize the current studies regarding metabolite production from *C. militaris* relative to their biological functions. We combined findings of metabolite production to correlate with different fermentation modes to obtain a full view of production processes used to yield the product. While research on *C. militaris* fermentation is not uncommon to date, its high value still highlights the importance of developing more modern fermentation processes for industrial production.

Keywords: *Cordyceps militaris*; extracellular polysaccharide; bioactivity; fermentation mode; metabolite

1. Introduction

Cordyceps militaris, which is called Yong Chong Cao in Chinese, is a type of entomopathogenic fungus that belongs to the *Cordyceps* genus in the phylum Ascomycota and family Cordycipitaceae [1]. It is most abundant in humid temperate and tropical regions, and is widely distributed throughout North America, Europe, and East and Southeast Asia [2]. *Cordyceps militaris* grows parasitically on the larvae and pupae of arthropods, mainly lepidopteron and coleopteran species. As the cultivation period progresses, the spores of *C. militaris* develop into hyphae, gradually growing into club-shaped, orange fruiting bodies measuring 2–5 cm in height. The morphology of mature *C. militaris* consists of two parts, including the stalk (the grass part, also known as the fruiting body) and the sclerotium (the corpse part of the insect) [3]. The basal medium for cultivating *C. militaris* typically consisted of 40 g of glucose (or other sugars), 10 g of peptone, 5 g of yeast extract, 1 g of K₂HPO₄, 1 g of KCl, 1 g of MgSO₄·7H₂O, and 2 g of adenosine per liter [4], without

pH control [5]. Niketan and Lakshmi [6] discovered that under the culture conditions of 20 °C temperature, 2.5% (*w/v*) glucose, and 0.8% (*w/v*) yeast extract, the maximal yield of the mycelial biomass reached 547 ± 2.09 mg/100 mL. This yield was 1.95-fold higher than that obtained in the basal medium.

All *Cordyceps* species share a mechanism of invading and growing on their hosts to maintain their life cycles [7], but the main difference occurs in the hosts they infect [8]. The majority of known species in the *Cordyceps* genus infect insects and other arthropods, with a few parasitizing fungi and plants. Most *Cordyceps* species exhibit host specificity and are known to infect particular or closely related host species, with only a few having the ability to infect a wide range of hosts [9]. During the parasitic process, *Cordyceps* has developed a complex survival mechanism to facilitate its attachment to a host and avoid clearance by the host's immune system. For this mechanism, *Cordyceps* produces several unique secondary metabolites that aid in its parasitism and reproduction on its hosts [7], showing great promise as potential sources of new drugs.

The *Cordyceps* genus has a history of several hundred years of use as a traditional Chinese medicine, with *C. sinensis* being the species with the longest history of use, exhibiting a wide range of biological activities. At present, the main source of *C. sinensis* is wild collection, with its rarity and high price are due to its wild characteristics. Although artificial cultivation has been achieved in recent years, production costs remain high, and cultivation times are long. In contrast, artificial cultivation of *C. militaris* yields lower production costs and shorter cultivation times compared to *C. sinensis*. *Cordyceps militaris* also has a similar chemical composition and comparable medicinal properties to those of *C. sinensis*, and even contains higher levels of certain bioactive components. Therefore, it is regarded as a promising substitute for *C. sinensis*. Figure 1 displays both natural and cultured *C. militaris* for comparison [10].

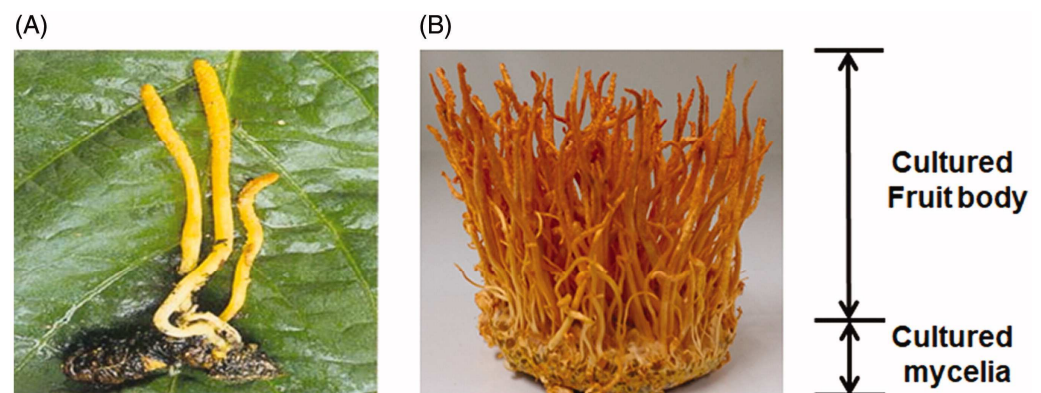


Figure 1. Natural (A) and cultivated (B) *C. militaris*. [7].

2. Biological Compounds of *C. militaris*

The bioactivities of *C. militaris* are significantly important for both medical and biological applications. Table 1 summarizes the physiological functions of various bioactive compounds extracted from *C. militaris*, which include cordycepin [11], D-mannitol [12], adenosine [13,14], polysaccharides [15], gamma aminobutyric acid (GABA) [16,17], lovastatin [16,17], and carotenoids [18,19], and these are further categorized and explained in the following paragraphs.

Table 1. Bioactive components of *Cordyceps militaris* and their biological activities.

Bioactive Compound	Biological Activity	References
	Antioxidant	[20–22]
	Anti-inflammatory	[23]
	Antiviral	[24]
	Antitumor	[24]
	Hypolipidemic	[25]
	Hypoglycemic	[26]
D-Mannitol	Antioxidant	[27]
	Anti-inflammatory	[28]
	Antitumor	[29]
	Hypolipidemic	[30]
	Hepatoprotective	[31]
	Antioxidant	[32,33]
	Anti-inflammatory	[34,35]
	Antiaging	[36,37]
	Anti-stress	[38]
	Cytoprotective	[39]
	Radioprotective	[40]
Adenosine	Antioxidant	[41,42]
	Anti-inflammatory	[43]
	Antitumor	[44]
	Anti-ischemic	[45]
	Antiplatelet	[46,47]
	Anti-hair loss	[48]
	Antioxidant	[49,50]
	Anti-inflammatory	[15]
	Antiviral	[51–53]
	Antiaging	[54]
	Antitumor	[55–57]
Gamma-aminobutyric acid	Antioxidant	[58,59]
	Anti-inflammatory	[60]
	Stress regulation	[61]
	Sleep-wake regulation	[61]
	Antihypertensive	[62]
	Antidiabetic	[63]
	Antioxidant	[64,65]
	Anti-inflammatory	[66,67]
	Antiviral	[68]
	Antitumor	[68]
	Antithrombotic	[69]
	Antiatherosclerotic	[70]
Carotenoids	Antioxidant	[71,72]
	Anti-inflammatory	[73,74]
	Antiaging	[73,74]
	Antimicrobial	[75,76]
	Antiapoptotic	[77]

2.1. Cordycepin

The molecular formula of cordycepin is $C_{10}H_{13}N_5O_3$, with a molecular weight of 251. In 1950, Cunningham et al. [11] discovered 3-deoxyadenosine in the fermentation broth of *C. militaris* and named it cordycepin. Cordycepin is an adenosine derivative, but it lacks a hydroxyl group at the 3' position of ribose. It was shown to possess diverse pharmacological properties, in addition to its well-known antioxidant effects [20–22], including antiviral activity [24], inhibition of lipopolysaccharide (LPS)-induced inflammation [57], reduction of blood lipids [78], inhibition of platelet aggregation [79], and induction of apoptosis in neuroblastoma and melanoma cells [80].

Cordycepin exhibits significant antioxidant effects by reducing the cellular content of malondialdehyde (MDA) and intracellular reactive oxygen species (ROS), and increasing

the activity of antioxidant enzymes such as superoxide dismutase (SOD) and glutathione peroxidase (GPx) in cells treated with 6-hydroxydopamine (6-OHDA) [22]. Cordycepin extraction using n-hexane led to a significant increase in activities of antioxidant enzymes. GPx was enhanced by 25.6% (5 μ M), 55.3% (10 μ M), and 98.4% (20 μ M), while SOD activity was, respectively, increased by 20.5%, 50.2%, and 77.4% [22].

Cordycepin has shown promise as a therapeutic agent for treating multiple inflammatory conditions, including rheumatoid arthritis [81], Parkinson's disease (PD) [82], multiple sclerosis (MS) [83], atherosclerosis [84], pneumonia [85], hepatitis [86], and atopic dermatitis (AD) [87,88]. For example, topical application of 2,4-dinitrofluorobenzene on the dorsal skin of mice induces AD-like lesions, which can be alleviated by cordycepin. This natural compound reduces serum levels of histamine, immunoglobulin E (IgE), and inflammatory cytokines; suppresses the infiltration of mast cells and eosinophils; and downregulates expressions of thymic stromal-derived lymphopoietin (TSLP), macrophage inflammatory protein (MIP)-2, intracellular cell adhesion molecule (ICAM)-1, thymus and activation-regulated chemokine (TARC), and C-C chemokine receptor (CCR)-3 [89].

Regarding the antiviral effect of cordycepin, it was found that cordycepin significantly inhibited the transfer of Epstein–Barr virus (EBV) from lymphoblastoid cell line (LCL)-EBV-green fluorescent protein (GFP) cells to AGS cells (a human gastric adenocarcinoma cell line), indicating a significant inhibition of EBV infection in gastric epithelial cells [24].

In terms of its antitumor effects, cordycepin significantly increased expressions of DNA methyl transferases (DNMTs), particularly DNMT3, leading to hypermethylation of the BCL7A tumor-suppressor gene in response to cordycepin treatment [24].

Cordycepin inhibits LPS-induced inflammation by suppressing the activation of nuclear factor (NF)- κ B, Akt, and p38 phosphorylation, resulting in the downregulation of inducible nitric oxide synthase (*i*NOS) and cyclooxygenase (COX)-2 gene expressions and subsequent inhibition of NO production [57]. It was shown to prevent hyperlipidemia by activating adenosine monophosphate-activated protein kinase (AMPK), demonstrating its potential as a functional food ingredient for managing lipid metabolism disorders [78].

In a dose-dependent manner, cordycepin inhibited platelet aggregation induced by thapsigargin and significantly decreased levels of $[Ca^{+}]_i$, which had been increased by thapsigargin (1 mM) or U46619 (a stable analog of the endoperoxide prostaglandin, H2) (3 mM). Furthermore, cordycepin increased cyclic guanosine monophosphate (cGMP) levels that had been reduced by thapsigargin [79].

The 50% inhibitory concentration (IC_{50}) values of cordycepin for human neuroblastoma SK-N-BE(2)-C and human melanoma SK-MEL-2 cells were 120 and 80 mM, respectively, indicating significant inhibition of their proliferation [80].

2.2. D-Mannitol

One of the most important products of *C. militaris* metabolism is D-mannitol, commonly known as cordycepic acid, which has a chemical formula of $C_6H_{14}O_6$, with a molecular weight of 182.172 g/mol, and is classified as a polyhydric alcohol (polyol) [16]. Mannitol has various medical applications due to its osmotic properties, and it is used in the food and pharmaceutical industries. It also stimulates the release of renal prostaglandins, resulting in renal vasodilation and increased urine flow through the tubules, which is thought to prevent renal injury by reducing tubular obstruction [90]. The biological activities of mannitol can be broadly classified into several categories, including antioxidant [27], anti-inflammatory, antitumor, hypolipidemic, and hepatoprotective properties.

As for the antioxidant ability of mannitol, it was shown to possess free radical-scavenging properties *in vitro*, protecting active substances such as hyaluronic acid from degradation by oxygen-derived free radicals. Moreover, it inhibits the release of ROS caused by the thermal- or photodecomposition of tetracyclines [91].

Mannitol was shown to have anti-inflammatory effects by inhibiting lipid peroxidation and formation of the NF- κ B complex [92].

Tumor-bearing animals treated with a high dose of cisplatin (CDDP; 9.5 mg/kg body weight (BW) i.v.) and mannitol survived longer than those given CDDP (7.0 or 9.5 mg/kg BW i.v.) alone, which suggests that mannitol can effectively enhance antitumor ability [29].

Mannitol (10 mg/kg BW) demonstrated hypolipidemic effects in hypercholesterolemic mice that were comparable to those of ezetimibe (10 mg/kg BW), without inducing liver damage or impacting high-density lipoprotein cholesterol (HDL-C) levels [30].

Mannitol administration has the potential to mitigate or alleviate liver injury caused by experimental obstructive jaundice, as elevated levels of platelet-activating factor (PAF) in both plasma and liver tissue were observed, which are thought to contribute to the progression of the injury [31].

2.3. Ergothioneine

Ergothioneine, with a molecular formula of $C_9H_{15}N_3O_2S$ and a molecular weight of 229.3 g/mol, is a water-soluble thiol compound. It is classified as a non-protein amino acid and is naturally synthesized by certain bacteria, plants, and fungi. Notably, mammals do not produce ergothioneine endogenously, underscoring its importance as an essential dietary nutrient. It has gained attention due to its identification as a biogenic key substrate of the organic cation transporter 1 (OCTN1), which has a protective role in monocytes and is associated with autoimmune disorders like rheumatoid arthritis and Crohn's disease [93]. *Cordyceps militaris* is a rich source of ergothioneine, as its fruiting bodies contain a concentration of 782.3 mg/kg dry weight (DW), and that of mycelium ranged up to 130.6 mg/kg DW [19]. Another study estimated a concentration of 409.8 mg/kg DW in its fruiting bodies [17].

Ergothioneine exhibits potent antioxidant properties, as it can scavenge hydroxyl radicals and inhibit their generation from hydrogen peroxide in a copper or iron ion-dependent manner. Additionally, it acts as an inhibitor of the copper ion-dependent oxidation of oxyhemoglobin, and of arachidonic acid peroxidation mediated by mixtures of myoglobin (or hemoglobin) and hydrogen peroxide. Furthermore, ergothioneine can protect alpha 1-antiproteinase from inactivation by hypochlorous acid due to its ability to effectively scavenge this reactive molecule [33].

Anti-inflammatory properties, including the modulation of interleukin (IL)-6, were attributed to ergothioneine. Furthermore, it may mitigate peroxyacetyl nitrate (PA)-induced cell death by diminishing the activity of the mitogen-activated protein kinase (MAPK) cascade [35].

The antiaging effect of ergothioneine is based on its ability to inhibit ultraviolet A (UVA)-induced activator protein (AP)-1 translocation, which in turn inhibits collagenolytic matrix metalloproteinase-1 activation and type I procollagen degradation [37].

Ergothioneine, along with its precursor, trimethyl histidine, and trehalose are considered anti-stress compounds that are maintained at high levels during starvation-mediated fasting [38].

Through activation of Nrf2 and FoxO3, ergothioneine exhibited cytoprotective effects, and FoxO3 appeared to play a role in erythroid proliferation and differentiation [94].

The radioprotective effect of ergothioneine is attributed to the reduction of ergothioneine disulfide, which is formed by irradiation-generated hydrogen peroxide, to ergothioneine in a neutral solution. Subsequently, the regenerated ergothioneine can again react with hydrogen peroxide [40].

2.4. Adenosine

Adenosine, with a molecular formula of $C_{10}H_{13}N_5O_4$ and a molecular weight of 267.2413 g/mol, is a nucleoside formed by the bonding of ribose and adenine through β -N9-glycosidic bonds [95]. It is identified as one of the primary active components in *C. militaris*, alongside other nucleosides like guanosine, cytidine, uridine, adenine, and uracil [96]. Its efficacy in treating chronic heart disease and modulating the release of neurotransmitters in the central nervous system was demonstrated by numerous studies [95,97,98].

The activation of cellular antioxidant enzymes is a novel mechanism by which adenosine exerts cytoprotective to the ischemic cell injuries [42].

Adenosine inhibits some functions of neutrophils, which are a type of inflammatory cell. By occupying specific adenosine receptors, it modulates leukocyte function through a novel mechanism that “uncouples” chemoattractant receptors from their stimulus-transduction proteins. Adenosine and its agonists are therefore excellent candidates for the development of anti-inflammatory agents [99].

Adenosine accumulates during tumor hypoxia and acts as a strong immunosuppressive agent by modulating the immune system. This can inhibit the antitumor immune responses elicited by standard radiotherapy, presenting ADO-mediated mechanisms that may thwart such responses [100].

The anti-ischemic effect exerted via the development of collateral circulation, at the heart and the brain level, is a therapeutically relevant possibility raised by the available evidence linking adenosine to angiogenesis [101].

Adenosine is involved in various pharmacological actions related to cardiovascular therapies. A_{2A} receptor agonists are being developed by the pharmaceutical industry for this purpose, and methotrexate, an atheroprotective and anti-inflammatory drug, exerts its effects through the release of adenosine and the activation of the A_{2A} receptor. Additionally, antiplatelet agents reduce platelet activation and adhesion and inhibit thrombotic occlusion of the atherosclerotic arteries by blocking adenosine diphosphate-mediated effects on the P₂Y₁₂ receptor [47,102–104].

By modulating the activity of Gsk3 β through adenosine receptor activation, adenosine stimulates the Wnt/ β -catenin signaling pathway. The hair growth-promoting effect of adenosine is attributed to the activation of the G α s/cAMP/protein kinase A (PKA)/mammalian target of the rapamycin (mTOR) cascade, which plays a pivotal role in adenosine-mediated Wnt pathway activation as the underlying molecular mechanism [48].

2.5. Polysaccharides

Polysaccharides are some of the most abundant and important bioactive components in *Cordyceps*, exhibiting a wide range of pharmacological activities and applications, including antioxidant [105], anti-inflammatory [106], antiviral [107], antiaging [108], and antitumor [107] effects. Polysaccharides can be extracted and isolated from *Cordyceps* fruiting bodies, mycelia, and fermentation broth, with different physicochemical properties, making them a targeted product for the development and quality control of *Cordyceps*-related health supplements [108]. In recent years, the isolation, purification, structural identification, and biological activities of *Cordyceps* polysaccharides have been extensively studied. However, due to variations in raw materials, extraction methods, and purification procedures, the structures and biological activities of polysaccharides obtained from *Cordyceps* exhibit significant differences, making it difficult to establish a correlation between structure, chain conformation, and biological activity. Therefore, it is necessary to develop a standardized preparation method for *Cordyceps* polysaccharides.

Microbial polysaccharides can be categorized based on their location within the cell, as follows:

- Structural polysaccharides, such as teichoic acids, LPSs, and peptidoglycans, form the cell wall to provide protection and maintain the microorganism's structure.
- Intracellular polysaccharides, also known as cytosolic polysaccharides, serve as a source of energy and carbon for the cell.
- Exopolysaccharides (EPSs) are secreted into the extracellular medium either as a secondary metabolite or in the form of a biofilm to protect against environmental stress [109,110].

With respect to the antioxidant activities of polysaccharides from *C. militaris*, the presence of certain monosaccharides may have a significant impact. In particular, a purified polysaccharide consisting primarily of glucose, mannose, galactose, and arabinose in an

α -type glycosidic linkage was found to exhibit *in vitro* scavenging activities against DPPH, hydroxyl, and superoxide radicals [49,50].

CPS-1, a polysaccharide with a molecular weight of 2.3×10^4 Da, is composed of rhamnose, xylose, mannose, glucose, and galactose, and was shown to have anti-inflammatory and humoral immunity-reducing properties [15].

An acidic polysaccharide extracted from *C. militaris* cultivated on germinated soybeans was found to possess antiviral properties against the influenza A virus. This effect was believed to be partly attributable to its ability to modulate the immune response of macrophages. The polysaccharide was shown to reduce viral titers in both bronchoalveolar lavage fluid and lung tissues. In addition, it was observed to enhance the production of tumor necrosis factor (TNF)- α and interferon (INF)- γ *in vivo*, as well as to stimulate nitric oxide (NO) production and induce expressions of inducible NO synthase (iNOS) messenger (m)RNA and protein *in vitro*. Furthermore, the polysaccharide was shown to modulate mRNA expressions of IL-1 β , IL-6, IL-10, and TNF- α [53].

Cultivated fruiting bodies of *C. militaris* containing polysaccharides were found to have effects on the activities of catalase (CAT), superoxide dismutase (SOD), glutathione peroxidase (GPx), and anti-hydroxyl radicals when assayed *in vivo* using commercial monitoring kits. These polysaccharides inhibited the mitochondrial injury and swelling induced by Fe²⁺-L-cysteine in a concentration-dependent manner and had a significant superoxide anion scavenging effect. Additionally, they significantly increased the activities of CAT, SOD, GPx, and anti-hydroxyl radicals in mice liver. These findings indicated that polysaccharides extracted from *C. militaris* may have pharmaceutical value in protecting mitochondria, scavenging ROS, and increasing the activities of antioxidases, which could potentially contribute to antiaging effects [54].

By orally administering 20 mg/kg BW of the water extract of the fruiting body of *C. militaris* containing polysaccharides to mice, it was discovered that the extracts could enhance the secretion of IFN- γ by macrophages through IL-18. This suggests that *C. militaris* has potential use as an immune activator or antitumor agent [56,57].

2.6. GABA

GABA, short for γ -Aminobutyric acid, has a molecular formula of C₄H₉NO₂ and a molecular weight of 103.12 g/mol. It is classified as a non-protein amino acid and is synthesized within the human body from glutamic acid. Concentrations of GABA were determined to be 68.6–180.1 mg/kg and 756.30 mg/kg dry weight in the mycelium and fruiting bodies of *C. militaris*, respectively [17]. GABA exhibits various biological activities, such as antioxidative, anti-inflammatory, anticonvulsant, antihypertensive, and antidiabetic properties.

GABA, known for its antioxidant properties [59], helps plants cope with various environmental stresses by providing nicotinamide adenine dinucleotide (NADH) and/or succinic acid via GABA shunts in the tricarboxylic acid cycle [111]. In addition to maintaining hormone and mineral nutrition, GABA reduces lipid peroxidation and enhances stress resistance in plants [112].

In relation to the anti-inflammatory ability of GABA, it exerts potent protective effects in models of type 1 diabetes by acting on both T cells and macrophages [60]. Additionally, GABAergic agents directly act on antigen-presenting cells (APCs), reducing MAPK signals and attenuating the following adaptive inflammatory responses to myelin proteins [113].

GABAergic neurons and neurotransmitters modulate various brain circuits to regulate stress and anxiety responses, under both normal and pathological conditions [114]. They also modulate both rapid eye movement (REM) and non-REM, especially slow wave sleep (SWS), through corticomedullary pathways [115]. Additionally, GABAergic neurons and neurotransmitters regulate circadian rhythms by modulating the suprachiasmatic nuclei (SCN) [116].

By activating GABA receptors, central GABA has the potential to lower blood pressure and slow down the heart rate, making it a valuable therapeutic option for preventing and

treating cardiovascular diseases resulting from hypertension, particularly in patients with cardiac insufficiency [117].

GABA, an antidiabetic factor coexisting with insulin in pancreatic β -cells [118], exerts its effect by acting on beta cell proliferation and the immune system. In patients with type 2 diabetes, a significant decrease in circulating GABA concentrations was observed, indicating its potential therapeutic value for diabetes [119].

2.7. Lovastatin

Lovastatin, found in the fruiting bodies of *C. militaris* [17,120], is a type of statin that selectively blocks the synthesis of endogenous cholesterol by inhibiting the rate-limiting enzyme in cholesterol production. Lovastatin features a six-membered lactone ring alongside a hydroxyl group and a partially hydrogenated naphthalene, with a hydroxyl substituent esterified by a 2-methylbutyric acid residue. Its molecular formula is $C_{24}H_{36}O_5$, with a molecular weight of 404.54 g/mol. It is converted from a lactone to a hydroxy acid form during enzymatic reactions, and this form competitively blocks 3-hydroxymethylglutaryl-coenzyme A (HMG-CoA) [121]. The lovastatin concentration in the mycelium of *C. militaris* ranged from 37.7~57.3 mg/kg DW, which was lower than the concentration found in *C. sinensis* at 1365.3 mg/kg [17]. Fruiting bodies of *C. militaris* contain lovastatin at a concentration of 2.76 mg/kg. However, the fruiting bodies of *Hericium erinaceus* and *Ganoderma lucidum* exhibit some of the highest lovastatin contents, at 14.38 and 11.54 mg/kg, respectively [16]. Approved for treating hypercholesterolemia and reducing the risk of coronary heart disease, lovastatin also exhibits additional beneficial effects such as anti-inflammatory and antioxidant properties, as well as vascular endothelium protection [122].

Upon oral administration, lovastatin was found to reduce oxidative stress and modulate activities of antioxidant enzymes in the liver and heart tissues of H_2O_2 -treated rats, indicating its potential in mitigating the effects of oxidative stress [65].

Lovastatin inhibits the phosphatidylinositol 3-kinase (PI3K)/Akt/mTOR signaling pathway and downregulates histone deacetylase 1 (HDAC1) expression, which prevents cytosolic NF- κ B from entering the nuclei of LPS-stimulated RAW264.7 macrophages. Additionally, it reduces expressions of iNOS and TNF- α and induces an anti-inflammatory response [67].

By directly inhibiting hydroxymethylglutaryl coenzyme A reductase, lovastatin inhibits the prenylation pathways in cells. It was found to reduce respiratory syncytial virus (RSV) replication when administered within 24 h after infection, but not vaccinia virus replication. In mice, lovastatin also decreases virus-induced weight loss and illness, indicating its potential as a therapeutic agent for RSV infection [123].

Lovastatin was demonstrated to enhance the antitumor effects of cisplatin and TNF- α , as well as to increase apoptosis induced by cisplatin, 5-fluorouracil, and sulindac in colon cancer cells [124].

Lovastatin plays an important role in inducing the expression of thrombomodulin, a gene with antithrombotic properties. Moreover, it decreases the expressions of prothrombotic factors such as E-selectin, plasminogen activator inhibitor type 1 (PAI-1), vascular cell adhesion molecule (VCAM)-1, intracellular cell adhesion molecule (ICAM)-1, and endothelin [125] and enhances the expression of the endothelial (e)NOS gene. Studies suggested that endothelial NOS expression is closely linked to thrombomodulin expression [126].

Lovastatin is effective as an antiatherosclerotic drug and a cell-cycle blocker due to its ability to induce cell-cycle arrest or retardation in various cell types. This effect is not only limited to the G₁ phase but also affects the G₂/M transition. Upon incubation of epithelial PtK2, T24, HeLa, and fibroblastic L929 cells with 1.0~60.0 μ M lovastatin for 24~48 h, various mitotic perturbations were observed [127].

2.8. Carotenoids

Carotenoids, such as β -carotene, lycopene, lutein, and zeaxanthin, are responsible for the vivid yellow-orange hue of the fruiting bodies of various species of fungi, including

C. militaris, in which their presence was confirmed [18,128]. In the aqueous extract of *C. militaris*, β -carotene was found at a concentration of 24.51 mg/kg, while lycopene was detected at 3.42 mg/kg [88]. Cordyxanthins are a novel class of carotenoids found in the fruiting bodies of *C. militaris*. They show better solubility in water and have unique chemical structures when compared to traditional carotenoids. Four types of cordyxanthins, including cordyxanthin I, II, III, and IV, have been identified in fruiting bodies of *C. militaris*, with respective concentrations of 0.289, 0.235, 0.401, and 0.175 mg/g. Using β -carotene as a standard, total *Cordyceps* carotenoids were directly quantified at 447 nm.

Carotenoids are powerful antioxidants due to their ability to efficiently quench singlet oxygen ($^1\text{O}_2$) [129–132], which is attributed to their triplet energy levels being in close proximity to that of $^1\text{O}_2$ [133,134]. The $^1\text{O}_2$ -quenching process is particularly efficient for carotenoids that have 11 conjugated double bonds [135], with an approximate rate constant of $10^{10} \text{ M}^{-1} \cdot \text{s}^{-1}$. In addition to their antioxidant properties, carotenoids were observed to act as protective agents against ROS in eye-related disorders [136]. Lutein and zeaxanthin, two oxygen-containing carotenoids, are present in high concentrations in the eye lens and the macular region of the retina (yellow spot) [137].

Carotenoids possess anti-inflammatory properties by suppressing the activation of NF- κ B [138–141]. Apocarotenals, which are derivatives of carotenoids, possess electrophilic groups that can interact with cysteine residues of I κ B kinase (IKK) and NF- κ B subunits (p65), leading to inactivation of the NF- κ B pathway [142].

Most carotenoids have demonstrated antiaging effects. For instance, an animal study showed that lutein can extend the lifespan and reduce the mortality rate caused by hydrogen peroxide and paraquat in *Drosophila melanogaster* [143], and further study indicated that lutein (at 0.5, 1.5, 5, and 15 μM) can mitigate the age-related decline in human skin cells [144]. Astaxanthin, another carotenoid, can also prolong the lifespan in wild-type and long-lived mutant age-1 *Caenorhabditis elegans* when administered at a concentration of 0.1–1 mM [145]. That study suggested that carotenoids partially modulate insulin-like growth factor (IGF)-1 signaling by increasing *DAF-16* gene expression and decreasing the mitochondrial production of ROS. Beta-carotene, a commonly found carotenoid derivative, also exerts antiaging effects by downregulating the expression of lysine acetyltransferase 7 (*KAT7*), while *KAT7* is a histone acetyltransferase gene known to be a driver of senescence [146].

Some research examined the antimicrobial effects of carotenoid pigments in *Rhodotorula* against two food-borne bacterial strains, *Staphylococcus aureus* ATCC 25,923 and *Salmonella typhimurium* ST38 [147]. Another study indicated that a wide range of antimicrobial activities against *Escherichia coli*, *Staphylococcus aureus*, *Bacillus subtilis*, *Saccharomyces cerevisiae*, and *Rhizopus oryzae* were exhibited by the methanol extract of carotenoids, as evidenced by the disc diffusion method and minimum inhibitory concentration (MIC) determination, with inhibition zones ranging 8.97–19.47 mm and MICs ranging 18.75–140 $\mu\text{g/mL}$ [148].

Antiapoptotic properties were demonstrated in some carotenoids. Lycopene, for instance, is capable of preventing the loss of such antiapoptotic proteins as Bcl-2 and Bcl-xL [149,150], as well as inhibiting proapoptotic proteins like Bax under neurotoxic conditions [149,151]. Lutein, on the other hand, is effective in preventing the loss of Bcl-2 and Bcl-xL, the accumulation of Bax, and the activation of caspases-3 and -8 in cerebral ischemia and Parkinson's disease models [152–154].

3. Cultivation of *C. militaris*

Wild *C. militaris* is rarely found in nature due to its extreme growth habitat, i.e., high-elevation areas, and its host specificity for arthropod pupae [155]. Due to great demands for the *C. militaris* in the therapeutic and cosmeceutical industry, the scaled-up production of *C. militaris* has drawn the attention of scientists and business sectors [156]. Based on these needs, the artificial cultivation of *Cordyceps* was developed and optimized to fulfill such a demand–supply gap and to prevent the extinction of natural *Cordyceps* resulting from over-harvesting. In addition, the mycelial cultivation of *C. militaris* was also deemed

important due to its rapid cultivation time, ease of scaling-up production, and ability to rapidly obtain specific biological compounds.

3.1. Solid-State Fermentation

Solid-state fermentation is commonly used for the artificial cultivation of *C. militaris* on a solid substrate, comprised of rice or grains, to obtain fruiting bodies [157,158]. However, it requires several months for the cultivation to form fruiting bodies, and it is difficult to control the quality of the final product [159]. Moreover, the large-scale extraction of bioactive compounds from *C. militaris* fruiting bodies is usually time-consuming and labor-intensive [1]. As a result, solid-state fermentation is unsuitable for obtaining biological ingredients from the large-scale industrial production of *C. militaris*. However, the fruiting body as the traditional consumption type of *C. militaris* still possesses high economic value in specific consumer markets, such as those offering Chinese herbs and medicinal diets [159].

In solid-state fermentation, the substrate plays an important role in *C. militaris* cultivation and the production of specific ingredients. Many substrates can be used for the solid-state fermentation of *C. militaris*, including wheat, oats, and rice [160]. Among these substrates, rice, which presents various advantages such as abundant nutrients, including amino acids, multiple B vitamins, and mineral elements, is the most commonly used solid-state substrate for *C. militaris* growth. Additionally, because rice is a cash crop, production costs are also relatively low [49]. In addition, Xiao, et al. [161] inoculated *C. militaris* on chickpeas to develop a new nutritious food product, and the results showed that fermented chickpeas presented higher contents of crude proteins, true proteins, and essential amino acids. In addition to producing specific products, *C. militaris* fermentation can also be used to improve the flavor of tea [162] under solid-state fermentation.

3.2. Liquid Culture

Liquid culture is the other option for *C. militaris* cultivation. It is considered a better procedure for application in industry due to its shorter cultivation times, smaller space requirements, and higher productivities [163]. *Cordyceps militaris* is allowed to grow in liquid as mycelia, without forming fruiting bodies, and total cultivation times can decrease from 60 to 15 days [164]. Bioactive compounds are released from mycelia and accumulate in the culture broth to simplify the extraction process [156]. Liquid culture can be carried out without the need for light/dark treatment and regulation of a specific relative humidity [158]. Therefore, liquid culture is considered a promising *C. militaris* cultivation procedure for obtaining useful and potent substances for industrial-scale applications.

In a recent study, the liquid culture of *C. militaris* was divided into submerged fermentation and surface culture methods. Submerged culture continually provides shaking or agitation during fermentation in order to dissolve nutrients and oxygen in the liquid medium [159]. For surface culture, the system remains in a static condition after the seeding of the inoculum. In surface culture, a biofilm forms on the surface, and some mycelia precipitate to the bottom of the culture flask, while the byproducts are directly secreted and accumulate in the medium [165]. In terms of metabolite yields, higher production can be achieved than that obtained using submerged culture; however, the surface culture time is relatively longer. In addition, surface culture is not easy to perform on an industrial scale [156]. Submerged culture can be carried out in a fermentation tank, and the fermentation time is relatively short and controllable [166]. Consequently, the *C. militaris* industry has mainly focused on developing the submerged culture method.

4. Production Mode of *Cordyceps militaris* and Metabolite Yields

4.1. Fed-Batch Culture and Batch Culture

For batch culture, all of the medium and the seed culture are fed into the bioreactor during cultivation, and the product remains in the bioreactor until cultivation is complete [167]. The environment of the batch culture process changes greatly with time. In the

later stage of culture, the lack of nutrients or the accumulation of inhibitory metabolites often causes difficulty in regards to cell survival [168]. Compared to batch culture, the fed-batch mode, through the continuous addition of nutrients, results in an increase in the total liquid volume to accumulate higher cell densities, which therefore improves the efficiency of the production process [169,170].

Mao and Zhong [171] found that feeding a nitrogen source during fermentation significantly increased the production of cordycepin from 208.8 to 346.1 mg/L. Furthermore, when *C. militaris* is mutated by UV irradiation, there is an increase in its cordycepin production, with a maximum cordycepin production of about 445 mg/L [172].

4.2. Repeated-Batch Fermentation

Repeated-batch fermentation is an economical, environmentally friendly, and stable method of fermentation, which can even be used as a modified approach for targeted products during production [173]. A repeated batch replaces the fresh medium after collecting the product at the end of each batch. It is expected that repeated-batch culture can increase the cell growth rate, ensuring high cell productivity [174,175]. During repeated-batch fermentation, microbial cells are reutilized for subsequent fermentation runs. Several benefits can be derived from this; for instance, reusing microbial cells for subsequent fermentation runs can achieve higher initial cell concentrations, and shorter operation times are required [174,176].

A previous study reported that the biomass of *C. militaris* can maintain a production level of >85% of that of the initial cycle for at least four repetitions by shortening the period of the lag phase by repeated-batch fermentation [174]. Another study showed that exopolysaccharide (EPS) production of *C. militaris* could be enhanced to a maximum of 5.713 g/L, with a productivity of 476 mg/L/day in the second run [165]. Moreover, Zheng et al. [177] combined repeated-batch fermentation and two-stage foam fractionation to develop a strategy to improve EPS production. The foam system can also promote the EPS separation efficiency during fractionation. Table 2 below illustrates the fermentation modes, types, and metabolite yields of different microbiota strains.

Table 2. Fermentation modes, types, and metabolite yields.

Strain Name	Fermentation Mode	Fermentation Type	Metabolite Yield (g/L)	Reference
BCRC 33735	Repeated batch	Stirred tank bioreactor	EPS: 5.713	[165]
BCRC 33735	Batch culture	Shake flask	EPS: 2.27	[165]
BCRC 33735	Batch culture	Shake flask	EPS: 2.4	[178]
ATCC 26848	Batch culture	Shake flask	Cordycepin: 0.1548 EPS: 1.5	[179]
ATCC 26848	Batch culture (two stage *)	Shake flask	Cordycepin: 2.2145	[179]
SU5	Batch culture	Shake flask	EPS: 1.9	[180]
CICC 14015	Batch culture (two stage *)	Shake flask	EPS: 3.2	[181]
CGMCC2459	Batch culture	Shake flask	Cordycepin: 2.008	[182]
G-813	Batch culture	Shake flask	Cordycepin: 6.84	[1]
CICC14015	Repeated batch	Shake flask	EPS: 1.218	[177]
Non	Batch culture	Stirred tank bioreactor	Cordycepin: 0.208	[171]
Non	Fed batch	Stirred tank bioreactor	Cordycepin: 0.346	[171]

* The first stage included a shaking condition, and the second stage used a static culture. EPS, exopolysaccharide.

5. Conclusions

In summary, *C. militaris* offers a promising alternative to *C. sinensis* in traditional medicine due to its shorter cultivation time and simpler growth requirements. Through fermentation, it produces bioactive compounds like cordycepin, with diverse physiological functions. Liquid culture, including fed-batch and batch culture, shows advantages over solid-state fermentation, including shorter cultivation times and simplified extraction processes. Repeated-batch fermentation holds promise for maintaining high production levels over multiple cycles. Optimizing fermentation conditions, i.e., by the addition of

nitrogen sources or UV irradiation-induced mutations, can significantly increase the production of key bioactive compounds. Understanding the production modes and metabolite yields of *C. militaris* is crucial for realizing its full potential in medicinal and industrial applications. Continued research into optimizing fermentation processes and exploring novel cultivation techniques will further unlock the therapeutic and commercial value of this valuable fungus.

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